

Premier University, Department of CSE

8th Semester, Assignment, Fall 2025

Course Title: Machine Learning

Course Code: CSE 457

Course Outcome: CO2

Total Marks: 10

Problem Title: AI-Based Automated Defect Detection System for Manufacturing

Objective:

Design a computer vision-based machine learning system that can automatically detect, classify, and localize defects in manufacturing products using image data.

Problem Context:

A manufacturing company produces electronic components. Manual inspection is slow and error-prone. The company wants to automate defect detection using deep learning and computer vision techniques.

Given Dataset:

1. Image Data:

- High-resolution product images
- Multiple defect types (cracks, scratches, missing parts)

2. Annotation Data:

- Bounding boxes for defects
- Class labels for each defect

Tasks:

Task 1: Data Preprocessing

- Image resizing and normalization
- Data augmentation (rotation, flipping, noise)
- Train-validation-test split

Task 2: Model Development

- Implement CNN-based models
- Compare models: CNN, ResNet, YOLO
- Perform hyperparameter tuning

Task 3: Object Detection

- Detect and localize defects using bounding boxes
- Evaluate using IoU and mAP

Task 4: System Deployment

- Design real-time inspection system
- Integrate camera and edge device

Task 5: Performance Evaluation

- Accuracy, Precision, Recall
- Inference time

Constraints:

- Real-time processing (< 1 second per image)
- Limited hardware resources
- Class imbalance

Expected Deliverables:

- Preprocessing pipeline
- Trained detection model
- Evaluation report
- System architecture diagram